OSSE1091E Practical Final Project

April 7th, 2026

Due: April 16th, 2026 @ 23:59

Introduction

You will complete a series of server management tasks throughout this Final Project. The tasks will reflect the practical work you've accomplished over the term. All material covered during the term is relevant to the project. You will also be tasked with completing novel concepts that were not performed in any lab but are relevant to concepts learned.

Please read this entire document before beginning the project.

It is also required that you complete all outstanding work before starting this project.

Screenshot Instructions

All delivered work will be **in the form of screenshots**. These screenshots will prove the work was done in the correct way. Please observe the following criteria when providing screenshots:

I. User **ONLY** windows Ink tool (WindowsKey + Shift + S) to gather screenshots. You may crop and mark-up any screenshots be **please leave the screenshots in their native .PNG format**.

II. When gathering screenshots, you **MUST** name the screenshots **based on the name provided in this document.** All names are indicated in **PURPLE**. Please append your initials (**_XXX**) to the end of the name that is indicated. Example:

8_PWSHMEM02 → **8_PWSHMEM02_GGB**

III. There will be a quiz in Brightspace associated with this assignment. Each question in this assignment will have an associated question in that quiz. **You will submit your screenshots to the relevant question in *Practical Final Project* quiz in Brightspace**.

IV. You only need to take screenshots **where indicated**. Ensure your screenshots show that you did the work. What is in your screenshot is up to you, but the validity of the screenshot is up to me. The screenshot must prove you did what was asked in the question. **Screenshots that do not show the required work (or are not clear) will be assigned a grade of zero.**

V. **Ensure your screenshots are legible**. Do not over-crop or shrink your screenshots excessively. Screenshots that are not legible **may be graded at zero**.

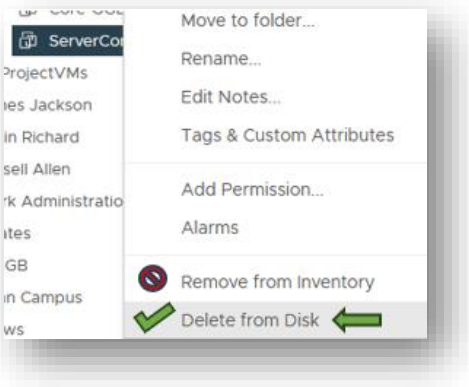
Preliminary Activity

If you have finished all outstanding work, please take a few minutes to delete all your VMs. **Again**: **you should complete all outstanding labs before doing this**. I would like to keep the ACE as environment tidy, and conserve storage where possible.

Important note: you may NOT use any of your VMs from the term. You MUST create new VMs. Using any pre-created or existing infrastructure will be considered **academic dishonesty**.

To delete your term VMs:

1. **In vSphere**: in your student container, delete Server1-XXX, Server2-XXX, Win11-XXX, Core-XXX, and any other VMs you have, **from Disk**. Create a folder named "Project" to hold your VMs.



Activity 1: Domain Controller

In vSphere, create domain controller using the "**Windows Server 2022 (WDSK)**" VM template.

1. In vSphere:
    a. Label the VM **DC01-XXX** (where XXX are your initials).
    b. Set the 2nd Network Adapter to your **isolated vSwitch**.
2. Once logged into the DC:
    a. Use the **first usable address** from your range of three addresses on **Ethernet0**.
    b. Use the address of **172.16.0.100** on **Ethernet1**. **Do not configure a Default gateway on this adapter**.
    c. Disable IPv6 on both Ethernet0 and Ethernet1.
    d. Hostname for this server should be **DC01-XXX** (where XXX is your initials).
    e. Set the time zone to **UTC-4** (Atlantic Canada).
    f. Turn OFF **Enhance Internet Explorer Configuration**.
    g. **Disable** the firewall **for all profiles**.
3. Install and configure **Active Directory Domain Services** (AD DS).
    a. Create a domain with the domain name of **<firstname>+<lastinitial>.exam**(example: J**effC.exam**).
4. Once the above is complete, **RDP** into DC01-XXX as the domain Administrator and provide a screenshot of the **Local Server page in Server manager**. Name the image "**1_DC01InitalConfig**".

Activity 2: Member Server 1

In vSphere, create member server using the "**Windows Server 2022 (WDSK)**" VM template.

1. In vSphere:
    a. Label the VM **MEM01-XXX** (where XXX are your initials).
2. Once Logged into the Server:
    a. Use the **second usable address** from your range of three addresses on **Ethernet0**.
    b. Disable IPv6 on both Ethernet0 and Ethernet1.
    c. Disable **Ethernet1**.
    d. Hostname for this server should be **MEM01-XXX** (where XXX is your initials).
    e. Set the time zone to **UTC-4** (Atlantic Canada).
    f. Turn OFF **Enhance Internet Explorer Configuration**.
    g. **Disable** the firewall for **all profiles**.
3. Join MEM01-XXX to your domain.
4. Once the above is complete, **RDP** into MEM01-XXX as the domain Administrator and provide a screenshot of the **Local Server page in Server manager**. Name the image "**2_MEM01InitalConfig**".

Activity 3: Member Server 2 (Core)

In vSphere create member server using the "**OSSE1091 Core (WDSK**) VM template.

1. In vSphere create a new VM in your folder:
    a. Label the VM **MEM02-XXX** (where XXX are your initials).

   b. Set the virtual switch to **AC_Wood-Fred-122**
2. Once Logged into the Server:
   a. Use the **third usable address** from your range of three addresses on **Ethernet0**.
   b. Hostname for this server should be **MEM02-XXX** (where XXX is your initials).
   c. Set the time zone to **UTC-4** (Atlantic Canada).
   d. Using Powershell, **disable IPv6 on Ethernet0**.
   e. **Disable** the firewall **for all profiles**.
3. Join MEM02-XXX to your domain.
4. Once the above is complete, RDP into MEM02-XXX as the domain Administrator and provide a screenshot showing the following.
   a. `IPCONFIG /ALL`. Name the image "**3_MEM02InitalConfig1**".
   b. `SCONFIG` main page. Name the image "**3_MEM02InitalConfig2**".
   c. Using **Powershell**, show that **IPv6 is Disabled**. Name the image "**3_MEM02InitalConfig3**".
   d. Using **Powershell**, show that the firewall is off for all profiles. Name the image "**3_MEM02InitalConfig4**".

Activity 4: Windows 11 Client

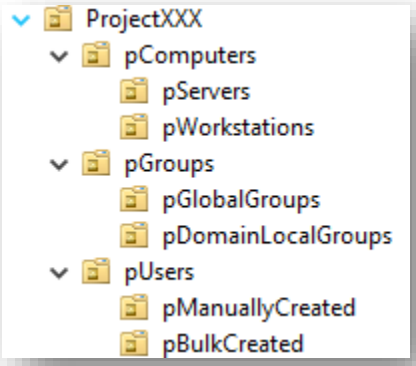
In vSphere, create Windows client using the "**Windows_11_(WSTK)**" VM template.

1. In vSphere:
   a. Label the VM **Win01-XXX** (where XXX are your initials).
   b. Set the Virtual Switch to your **isolated vSwitch**.
2. Once the Client boots to Windows:
   a. Create the initial account **EXLADMIN_XXX** (where XXX is your initials) during OOBE.
   b. Complete the OOBE.
   c. Leave **Ethernet0** set to **DHCP**.
   d. **Disable** IPv6 on **Ethernet0**.
   e. Hostname for this client should be **Win01-XXX** (where XXX is your initials).
   f. Set the time zone to **UTC-4** (Atlantic Canada).
   g. **Leave the Firewall Turned ON.**
3. **Do NOT** join Win01-XXX to your domain currently.
4. **Do NOT** configure RDP currently.
5. Once the above is complete, use the **vSphere web console** to log in as the **EXLADMIN_XXX**. Provide a screenshot showing the following configuration items:
   a. A `WHOAMI` in CMD
   b. An `IPCONFIG /ALL` in CMD.
   c. The time zone is configured correctly.

   Name the image "**4_Win01InitalConfig**".

Activity 5: Organization Unit Structure

1. Create the following OU structure **using Powershell** (where XXX are your initials), in the root of the domain.



2. Take a screenshot of the commands executing successfully. Name the image "**5_PWSHOUStructure**".
3. Using **Powershell**, move all **member servers** to the "**ProjectXXX\pComputers\pServers OU**".
4. Take a screenshot of the commands executing successfully. Name the image "**5_PWSHMemMove**".

Activity 6: DHCP for the isolated network.

1. On your **domain controller**, install the **DHCP role**.
2. **Unbind** the DHCP server from **Ethernet0**. This is **important**. I do not want DHCP offered on the 10.157.126.0/23 network.
3. **Once you have unbound Ethernet0**:
    a. Create a scope for the **172.16.0.0/24** network.
    b. Name the scope "**IsolateNetScope_XXX**" (where XXX is your initials).
    c. Configure the scope appropriately for the above network. Include all usable addresses in the scope.
    d. Exclude leases for the following:
        i. Exclude the first 50 addresses of the subnet.
        ii. Exclude the address 172.16.0.100.
        iii. Exclude the last 10 addresses of the subnet.
    e. The lease duration should be set to ½ of a day.
    f. The DHCP lease should set the client's **default gateway** to 172.16.0.1.
    g. The DHCP lease should set the client's **Preferred DNS** server to DC01-XXX.
    h. Take a screenshot of the **address pool**, configured correctly. Name the image "**6_DHCPPool**".
    i. Take a screenshot of the **scope options**, configured correctly. Name the image "**6_DHCPOptions**".
    j. Take a screenshot of Win01-XXX obtaining a lease in the allotted range. Name the screenshot "**6_WIN01DCHP1**".

   k. **Ensure you have the previous screenshot before continuing**. Set a DHCP reservation for Win01-XXX for the address **172.16.0.75**.
   l. Take a screenshot of Win01-XXX obtaining a lease from the reserved address. Show the "**Address Leases**" section of the DHCP server indicating that the **reservation is "Active"**. Name the screenshot "**6_WIN01DCHP2**".
4. Join Win01-XXX to the domain. Once joined to the domain, move its computer account to the "**ProjectXXX\pComputers\pWorkstations**" OU.

Activity 7: RDP GPO.

1. Create a group structure that **follows Microsoft's best practices**, to allow any user who is a member of the "correct group" to RDP into any computer in the domain **except** the domain controller.
2. Put any Global Groups in "**ProjectXXX\pGroups\pGlobalGroups**" and any Domain Local Groups in "**ProjectXXX\pGroups\pDomainLocalGroups**".
3. Create Group Policy Object **with the appropriate settings to allow RDP**. **Link this GPO to the appropriate OU** to allow your **servers and workstations** to be RDP'd into by any user specified in your RDP group(s).
4. Take a screenshot of your RDP Groups. Name the image "**7_RDPGroups**".
5. Take a screenshot of the GPO settings and its link. Ensure that **all relevant GPO settings are visible in the screenshot**. Name the image "**7_RDPGPO**".

Activity 8: RDP into Win01-XXX from DC01-XXX.

1. With the Win01-XXX firewall **ON**, RDP **from DC01-XXX to Win01-XXX**.
2. Once RDP'd into Win01-XXX, take a screenshot showing the following.
   a. You are connected VIA RDP.
   b. The firewall is on.
   c. `IPCONFIG /ALL` in CMD.

   Name the image "**8_WIN01RDP1**".
3. While RDP'd, provide a screenshot of the "Remote" tab in advanced system properties. Name the image "**8_WIN01RDP2**".
4. While RDP'd, use `GPRESULT` to produce a report for the polices applied **to the computer**. Show that the settings from your RDP GPO have been applied successfully. Name the image "**8_WIN01RDP3**".

Activity 8: Install Powershell Core 7.6.X.

1. Install Powershell Core **on all Servers and Clients**.
2. To install Powershell core on DC01-XXX:
   a. Create a shared folder called "**DomainFiles**" on **DC01-XXX**. Assign **Everyone → Full Control** share permission.
   b. Put the installer **.MSI** for Powershell core in this shared folder.
   c. On **MEM02-XXX**, using **CMD or Powershell**, map the "**DomainFiles**" share folder to the **S:** drive letter.
3. Install Powershell core to MEM02-XXX **using the DomainFiles share**.

4. Install PWSH Core to Win01-XXX **using the DomainFiles share**.
5. Provide a screenshot of PWSH Core 7.6.x installed on MEM02-XXX. Show the mapped Shared folder in your screenshot. Name the image "**8_PWSHMEM02**".
6. Enable **PSRemoting** on all VMs. Take a screenshot showing PSRemoting is enabled on Win01-XXX. Name the image "**8_PWSHWIN01**".

Activity 9: Personal Domain Administrator Account.

1. Using AD UC, create a Domain Administrator account for yourself.
    a. Name the account **EXDADMIN_XXX** (where XXX are your initials).
    b. First Name: <your name>
    c. Last Name: <your name>
    d. Full Name: EXDADMIN_XXX
    e. Display Name: <your first and last name>
    f. Set the password of the account to "never expire".
    g. Move the user to "**ProjectXXX\pUsers**".
2. Take a screenshot Showing that the account is a domain administrator and shows the above account properties configured correctly. Name the image "**9_DADMIN**".

Activity 10: New Users (AD UC).

1. On **MEM01-XXX**, create a folder on your server for user home directories.
    a. Call the folder **C:\homefolders**.
    b. Share the folder as "**WorkFolders**".
    c. Give **everyone** the **Full Control** share permission.
2. Create a **user template**. This template will be used to create all "manually created" users. Give the template the following properties:
    a. Create the template in **ProjectXXX\pUsers**.
    b. Give the account a full name of "**_manUserTemplate**".
    c. Give the account a logon of "**manUserTemp**".
    d. Give the account a password of **"P@$$w0rd"**.
    e. Set the "User must change password at next logon" as **enabled**.
    f. **Disable** the account.
3. Take a screenshot showing the **summary page** of the Template account during creation. Name the image "**10_TemplateAccount1**".
4. The template account should be configured with a home folder in the "**WorkFolders**" share on **MEM01-XXX**. This should be mapped to the user's **H:** drive at logon.
    a. Take a screenshot of this configuration. Name the image "**10_TemplateAccount2**".
5. Ensure that the template account **is allowed to RDP** so that users created from this template can also RDP.
    a. Take a screenshot of this configuration. Name the image "**10_TemplateAccount3**".
6. Using the template account, create **five new users**.
    a. Each user should have a full name of manUser1-5 (manUser1, manUser2, etc).
    b. Create a First and Last name for each user (your choice).
    c. UPN for each user should be **manUser#@<domain>.exam**.
    d. **SamAccountName** (legacy name) should be the same as the logon name.

   e. Provide a screenshot **of the summary page** (during creation) for at least one of the users created with the template. Name the image "**10_manUsers**".
7. Move all five of the manually created users to the "**ProjectXXX\pUsers\pManuallyCreated**" OU.
8. **Using AD UC**, enable all five users ***at the same time***. Take a screenshot of the window while doing this. Name the image "**10_manUserEnable**".

Activity 11: New Users (PWSH).

You will create a small Powershell script to create 5 users in bulk.

1. The script will create **five new users**.
2. Create a **CSV file** in excel with the user information to be imported by Powershell.
3. The script will use the CSV file to import most of the information for each account. The following information about each user **should be derived from the CSV**.
   a. **FirstName** (your choice)
   b. **LastName** (your choice)
   c. **UPN** of **bulkUser#@<domain>.exam** (where # is the 1-5).
   d. **SamAccountname** of bulkUser# (where # is the 1-5).
   e. The **Name** parameter is the same name as the SamAccountName
   f. **Home folder** should be name as the users SamAccountName and created in the "**WorkFolders**" share on MEM01-XXX.
   g. Users should be created in "**Project-XXX\pUsers\pBulkCreated**" OU during creation.
4. The script will use a **ForEach-Object** loop to process the CSV.
5. The user should be **flagged to change their password at the next logon** at **creation**.
6. The script, upon completion, will display the **following properties of all five users in a table**:
   a. GivenName
   b. Surname
   c. SamAccountName
   d. UserPrincipalName
   e. Enabled status.
7. **Only comment the script with your name, group number, and course number at the top of the script.**
8. **Do NOT use any code or add any functionality that is outside of that requested above**.
9. There is **NO SCREENSHOT** for this activity. I want you to **submit the script with your Project**. I will validate the script manually.
10. **If you cannot get the script to work**, create the bulkUsers1-5 manually and move them to the "**Project-XXX\pUsers\pBulkCreated**" OU.

Activity 12: Role installation

Install the following roles on the indicated servers.

1. On **MEM01-XXX**:
   a. Install the **Printer Management** role.
   b. Take a screenshot showing the above role installed. Name the image "**12_MEM01Role**".
2. Using **Server Manager or PWSH**, install the following role on MEM02-XXX:

   a. Install the **DNS** role.
   b. **Using a remote Powershell from session DC01-XXX** show that the DNS role is installed on MEM02-XXX. Take a screenshot showing the above role installed using remote Powershell. Name the image "**12_MEM02Role**".

Activity 13: Printer Server configuration

MEM01-XXX will be our simulated Print Server with internet printing.

1. On MEM01-XXX in the print management console install the **HP Universal PCL 6** print driver.
2. During Installation of the print driver, install a printer using **LPT1**:
   a. Name the printer "**Project Printer 1 XXX**" (where XXX are your initials).
   b. Share the Printer with the same name.
   c. List the printer in the directory.
3. In the Print Management console, add a second printer on LPT2:
   a. Name the printer "**Project Printer 2 (after hours) XXX**" (where XXX are your initials).
   b. Share the Printer with the same name.
   c. List the printer in the directory.
   d. Set this printer's to **only be available from 5:00 PM to 8:00 AM**.
   e. Take a screenshot of this printer's schedule. Name the screenshot "**13_PrintSched**".
4. In the Print Management console, add a third printer on LPT3:
   a. Name the printer "**Project Printer 3 (pooled) XXX**" (where XXX are your initials).
   b. Share the Printer with the same name.
   c. List the printer in the directory.
   d. Configure "**Project Printer 3 (pooled) XXX"** as a pooled printer (Pooled with LPT1, LPT2 and LPT3).
   e. Take a screenshot of this printer's pool. Name the screenshot "**13_PrintPool**".
   f. Add **EXDADMIN_XXX** to the ACL of this printer with the "**Print**", "**Manage this Printer**", and "**Manage Documents**" permissions, explicitly.
   g. Take a screenshot of this printer's permissions. Name the screenshot "**13_PrintPerms**".
5. Take a screenshot in AD UC or Powershell showing that all three printers are listed in the directory. Name the screenshot "**13_PrintListed**".

Activity 14: DNS Configuration

1. Using remote management tools or PWSH, configure the DNS server on DC01-XX and MEM02-XXX:
   a. Authorize MEM02-XXX as a **Name Server** and create the NS record for MEM02-XXX.
   b. Authorize MEM02-XXX to accept **Zone Transfers**.
2. Provide a screenshot showing the **NS** record for MEM02-XXX and that it is authorized to receive zone transfers. Name the image "**14_DNSNSSever**".
3. On **MEM02-XXX** create a **secondary standard DNS zone for <domain>.exam**.
   a. Perform a Zone Transfer from the Master DNS server.
   b. Take a screenshot showing that the transfer was successful. Name the image "**14_DNSSecondary**".

4. On **DC01-XXX** Create a Reverse Lookup zone for the 10.157.0.0/16 subnet on the Primary DNS Server.
    a. Take a screenshot showing the PTR records for **DC01-XXX**, **MEM01-XXX**, and **MEM02-XXX**. Name the image "**14_RLZ**".
5. Create a **CNAME** record for **MEM01-XXX**.
    a. The CNAME record's FQDN should be "**PrintServer.<domain>.exam**".
    b. **Using** NSLOOKUP, demonstrate that the alias record resolves to the IP address of MEM01-XXX. Name the image "**14_CNAME**".

Activity 15: GPO configuration.

1. Edit the Default Domain policy.
    a. Modify the domain password policy to reflect the following:
        i. 6 passwords remembered.
        ii. Maximum password age of 90 days.
        iii. Minimum Password length of eight characters.
        iv. Take a screenshot showing the password policy configured correctly. Name the image "**15_PassPolicy**".
    b. Modify the lockout Policy to reflect the following:
        i. Lockout duration of two minutes.
        ii. Twelve password attempts before locking the account.
        iii. Reset the lockout count after one minute.
        iv. Take a screenshot showing the lockout policy configured correctly. Name the image "**15_LockoutPolicy**".
    c. Configure the Default Domain policy so that it always has the highest precedence on **all child OUs**.
        i. Take a screenshot showing this configuration. Name the image "**15_DomainGPOWins**".
2. Create a new GPO and link it to the **"ProjectXXX\pUsers"** OU.
    a. Call the policy "**PooledPrinter**".
    b. Configure the policy so that the "**Project Printer 3 (pooled) XXX**" is deployed to all users in the "**ProjectXXX\pUsers\pManuallyCreated**" OU, but NOT the users in the "**ProjectXXX\pUsers\pBulkCreated**" OU.
    c. Take a screenshot showing this configuration. Name the image "**15_GPOPooledPrinters**".
3. Create a new GPO and link it to the **"ProjectXXX\pUsers"** OU.
    a. Call the policy "**AfterHoursPrinter**".
    b. Configure the policy so that the "**Project Printer 2 (after hours) XXX**" is deployed to all users in the "**ProjectXXX\pUsers**" OU except **bulkUser5** (specifically).
    c. Take a screenshot showing this configuration. Name the image "**15_GPOnoBulkUser5**".

Activity 16: AD AC.

1. Turn on the **Active Directory Recycle Bin**.
    a. Take a screenshot showing you've done the above. Name the image "**16_Recycle**".
2. Configure a **fine-grained password policy** for **EXDADMIN_XXX**.

a. Call the policy "**EXDADMIN_FGPP**".
b. Give the policy the highest precedence (overriding any other policy).
c. Change the password length to 25 characters.
d. 25 remembered passwords.
e. Minimum Password age of one
f. Maximum password age of 60
g. Twelve failed password attempts.
h. Failed attempt reset counter of one minute.
i. A lockout duration of **indefinite**.
j. Take a screenshot showing you've done the above. Name the image "**16_FGPP**".

Marking Breakdown.

Num.	Graded Item	Points	
1	1_DC01InitalConfig	7	
2	2_MEM01InitalConfig	7	
3	3_MEM02InitalConfig1	1	
4	3_MEM02InitalConfig2	1	
5	3_MEM02InitalConfig3	1	
6	3_MEM02InitalConfig4	1	
7	4_Win01InitalConfig	3	
8	5_PWSHOUStructure	1	
9	5_PWSHMemMove	1	
10	6_DHCPPool	3	
11	6_DHCPOptions	2	
12	6_WIN01DCHP1	1	
13	6_WIN01DCHP2	1	
14	7_RDPGroups	2	
15	7_RDPGPO	2	
16	8_WIN01RDP1	2	
17	8_WIN01RDP2	1	
18	8_WIN01RDP3	1	
19	8_PWSHMEM02	2	
20	8_PWSHWIN01	1	
21	9_DADMIN	1	
22	10_TemplateAccount1	1	
23	10_TemplateAccount2	1	
24	10_TemplateAccount3	1	
25	10_manUsers	1	
26	10_manUserEnable	1	
27	BULK USER SCRIPT	4	
28	12_MEM01Role	2	
29	12_MEM02Role	2	

Num.	Graded Item	Points	
30	13_PrintSched	1	
31	13_PrintPool	1	
32	13_PrintPerms	1	
33	13_PrintListed	1	
34	14_DNSNSSever	2	
35	14_DNSSecondary	2	
36	14_RLZ	2	
37	14_CNAME	1	
38	15_PassPolicy	1	
39	15_LockoutPolicy	1	
40	15_DomainGPOWins	1	
41	15_GPOPooledPrinters	2	
42	15_GPOnoBulkUser5	2	
43	16_Recycle	1	
44	16_FGPP	1	

Total points	76